

# Security Operations Center (SOC) Practical Task Report

## Title

Security Monitoring, Log Analysis and Incident Response Practical Implementation

---

## Objective

The objective of this task is to understand the working of a Security Operations Center (SOC) by performing practical activities such as log monitoring, browser history analysis, and network traffic monitoring. The task also aims to develop knowledge about incident documentation and alert rule design used in real-world security operations.

---

## 1. Theoretical Knowledge

### SOC Fundamentals

A Security Operations Center (SOC) is responsible for continuous monitoring of organizational IT infrastructure to detect and respond to cyber threats. SOC analysts work in different tiers where Tier-1 handles alert monitoring, Tier-2 performs detailed investigation, and Tier-3 focuses on threat hunting and advanced incident response.

### Security Monitoring Basics

Security monitoring involves analyzing logs and network activities to detect anomalies and unauthorized access attempts. Security Information and Event Management (SIEM) tools help in correlating events and generating alerts. Important metrics include false positives and Mean Time to Detect (MTTD).

### Log Management

Log management includes collection, normalization, storage, and analysis of system logs. Common log sources include Windows Event Logs, Syslog from Linux systems, and web server access logs. Effective log analysis helps in early detection of cyber attacks.

### Security Tools Overview

Various tools are used in SOC environments such as Wireshark for network packet analysis, SIEM tools for centralized monitoring, intrusion detection systems for attack detection, and endpoint monitoring tools for system visibility.

**Basic Security Concepts**

The CIA triad includes Confidentiality, Integrity, and Availability which are essential principles of information security. A threat represents a potential attacker, a vulnerability is a system weakness, and risk is the likelihood of threat exploitation.

**Security Operations Workflow**

SOC workflow includes detection of alerts, triage based on severity, investigation using log correlation, and response actions such as containment and remediation.

**Incident Response Lifecycle**

Incident response follows stages including preparation, identification, containment, eradication, recovery, and lessons learned. Proper response ensures minimal operational impact during security incidents.

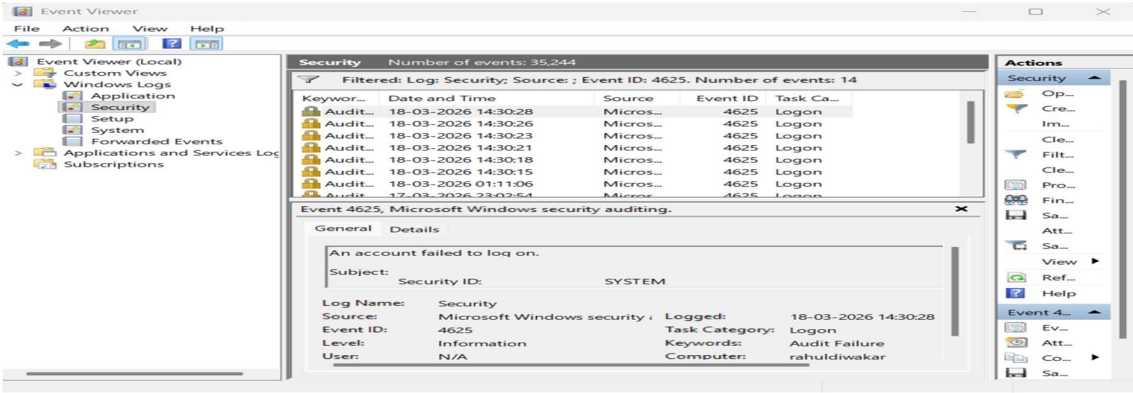
**Documentation Standards**

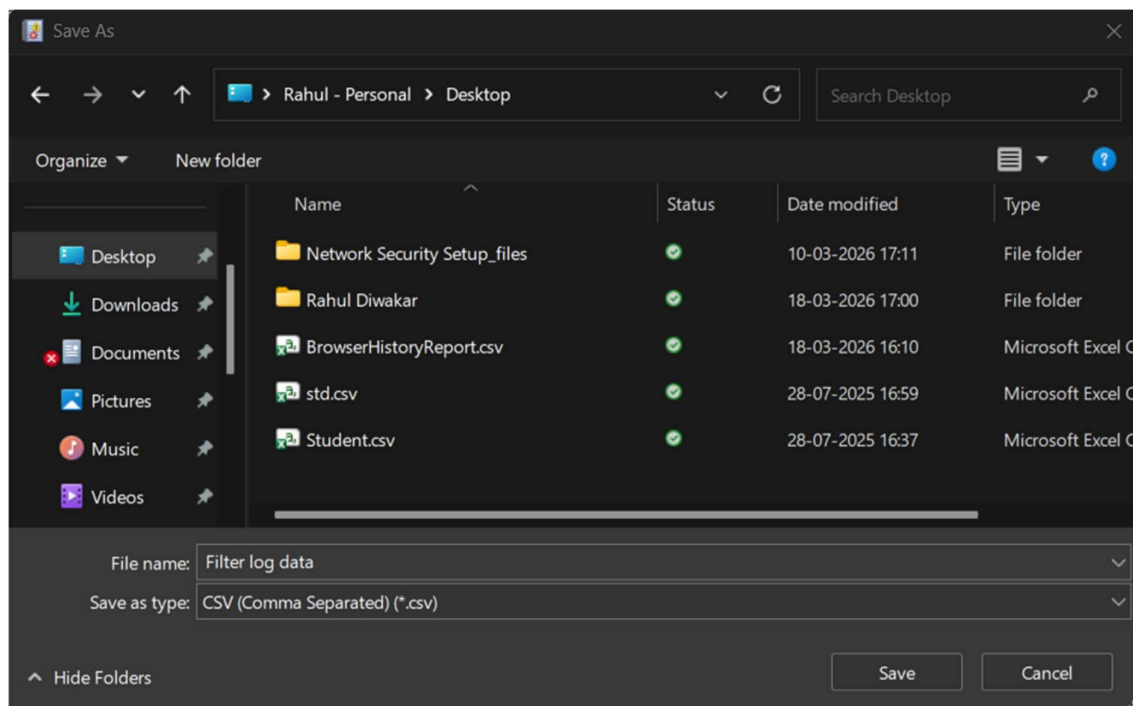
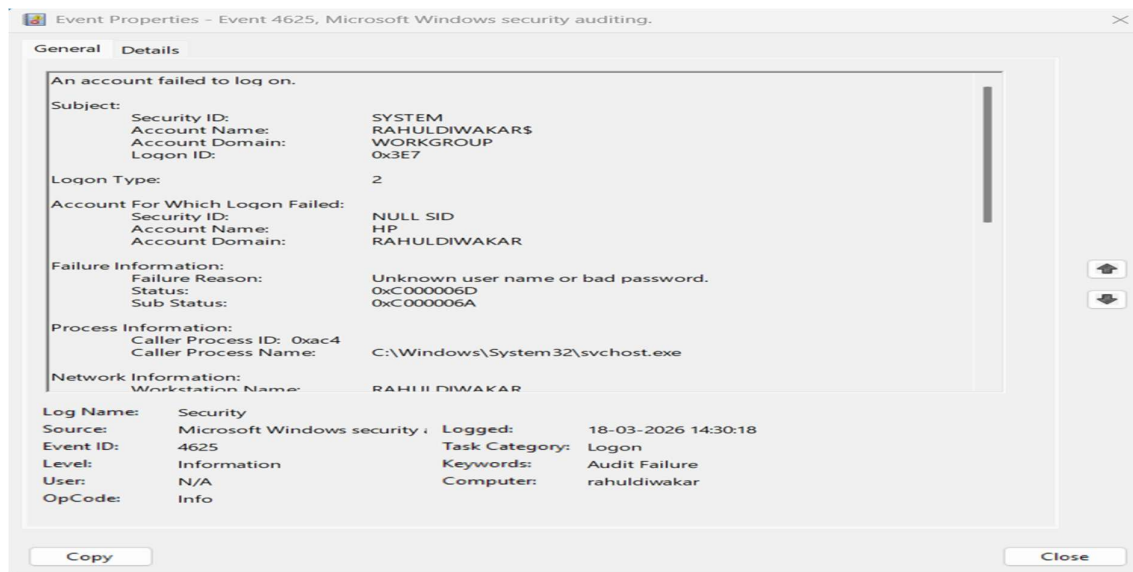
Security incidents must be documented using structured formats including incident reports, standard operating procedures (SOPs), and post-incident analysis reports.

**2. Practical Implementation**

**Log Analysis using Event Viewer**

Windows Event Viewer was used to monitor security logs. Event ID 4625 was filtered to detect failed login attempts. Multiple authentication failures were identified indicating potential brute-force activity. The filtered logs were exported for further analysis.





## Browser History Forensics

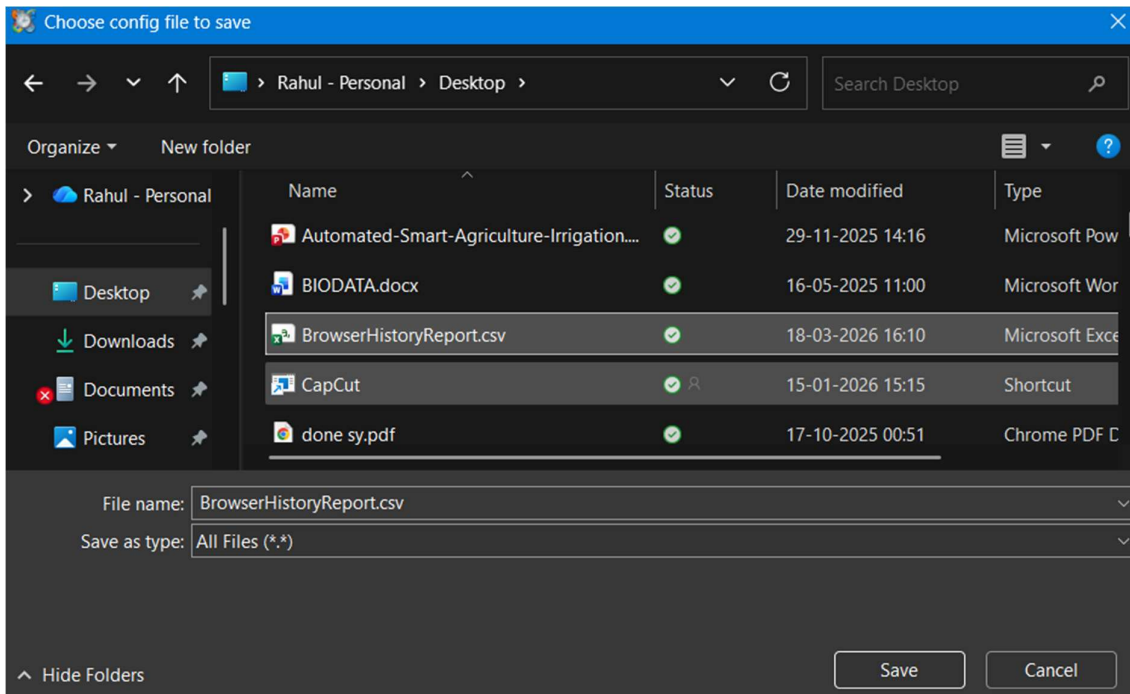
BrowserHistoryView tool was used to analyze web browsing activity. The tool displayed visited URLs along with timestamps and browser details. The browsing history was exported into CSV format as forensic evidence.

| BrowsingHistoryView         |                             |                     |             |              |            |                |                              |              |                 |            |
|-----------------------------|-----------------------------|---------------------|-------------|--------------|------------|----------------|------------------------------|--------------|-----------------|------------|
| File Edit View Options Help |                             |                     |             |              |            |                |                              |              |                 |            |
| URL                         | Title                       | Visit Time          | Visit Count | Visited From | Visit Type | Visit Duration | Web Browser                  | User Profile | Browser Profile | URL Length |
| file:///C:/Program%20...    | Wireshark 4.6.4 Release ... | 10-03-2026 14:16:10 | 1           |              |            |                | Edge (Chromium-based)        | HP           | Default         | 69         |
| file:///C:/Users/HP/Do...   |                             | 12-03-2026 16:36:47 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 181        |
| file:///C:/Users/HP/Do...   | (anonymous)                 | 12-03-2026 16:37:42 | 3           |              |            |                | Edge (Chromium-based)        | HP           | Default         | 181        |
| file:///C:/Users/HP/Do...   | (anonymous)                 | 12-03-2026 16:37:53 | 2           |              |            |                | Edge (Chromium-based)        | HP           | Default         | 201        |
| file:///C:/Users/HP/Do...   |                             | 12-03-2026 16:37:53 | 2           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 201        |
| file:///C:/Users/HP/Do...   |                             | 18-03-2026 00:41:22 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 213        |
| file:///C:/Users/HP/Do...   |                             | 12-03-2026 14:00:36 | 2           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 67         |
| file:///C:/Users/HP/Do...   | Network Security Setup      | 10-03-2026 17:22:23 | 1           |              |            |                | Edge (Chromium-based)        | HP           | Default         | 61         |
| file:///C:/Users/HP/Do...   |                             | 10-03-2026 17:22:08 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 61         |
| file:///C:/Users/HP/Do...   |                             | 12-03-2026 16:28:30 | 2           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 68         |
| file:///C:/Users/HP/Do...   |                             | 13-03-2026 17:10:36 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 53         |
| file:///C:/Users/HP/Do...   |                             | 13-03-2026 17:12:17 | 2           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 58         |
| file:///C:/Users/HP/Do...   |                             | 12-03-2026 16:44:40 | 3           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 51         |
| file:///C:/Users/HP/Do...   |                             | 12-03-2026 16:51:16 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 47         |
| file:///C:/Users/HP/Do...   | (anonymous)                 | 12-03-2026 16:49:29 | 1           |              |            |                | Edge (Chromium-based)        | HP           | Default         | 53         |
| file:///C:/Users/HP/Do...   |                             | 10-03-2026 17:33:05 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 45         |
| file:///C:/Users/HP/Do...   |                             | 10-03-2026 15:59:56 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 62         |
| file:///C:/Users/HP/On...   |                             | 18-03-2026 14:43:56 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 60         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:10:56 | 2           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 68         |
| file:///C:/Users/HP/On...   | Network Security Setup      | 10-03-2026 17:11:04 | 3           |              |            |                | Edge (Chromium-based)        | HP           | Default         | 68         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:08:38 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 84         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:06:59 | 2           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 87         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:06:51 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 101        |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:06:36 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 93         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:08:50 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 88         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:09:06 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 76         |
| file:///C:/Users/HP/On...   | Network Security Setup      | 10-03-2026 17:10:05 | 1           |              |            |                | Edge (Chromium-based)        | HP           | Default         | 84         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:09:51 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 84         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 17:06:22 | 4           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 82         |
| file:///C:/Users/HP/On...   |                             | 18-03-2026 02:31:18 | 3           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 89         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 14:26:08 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |
| file:///C:/Users/HP/On...   |                             | 10-03-2026 14:46:47 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |
| file:///C:/Users/HP/On...   |                             | 12-03-2026 13:24:17 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |
| file:///C:/Users/HP/On...   |                             | 12-03-2026 13:48:31 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |
| file:///C:/Users/HP/On...   |                             | 12-03-2026 13:58:31 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |
| file:///C:/Users/HP/On...   |                             | 13-03-2026 15:26:08 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |
| file:///C:/Users/HP/On...   |                             | 12-03-2026 16:07:34 | 2           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |
| file:///C:/Users/HP/On...   |                             | 12-03-2026 16:07:43 | 1           |              |            |                | Internet Explorer 10/11 / .. | HP           |                 | 86         |

555 Item(s), 1 Selected

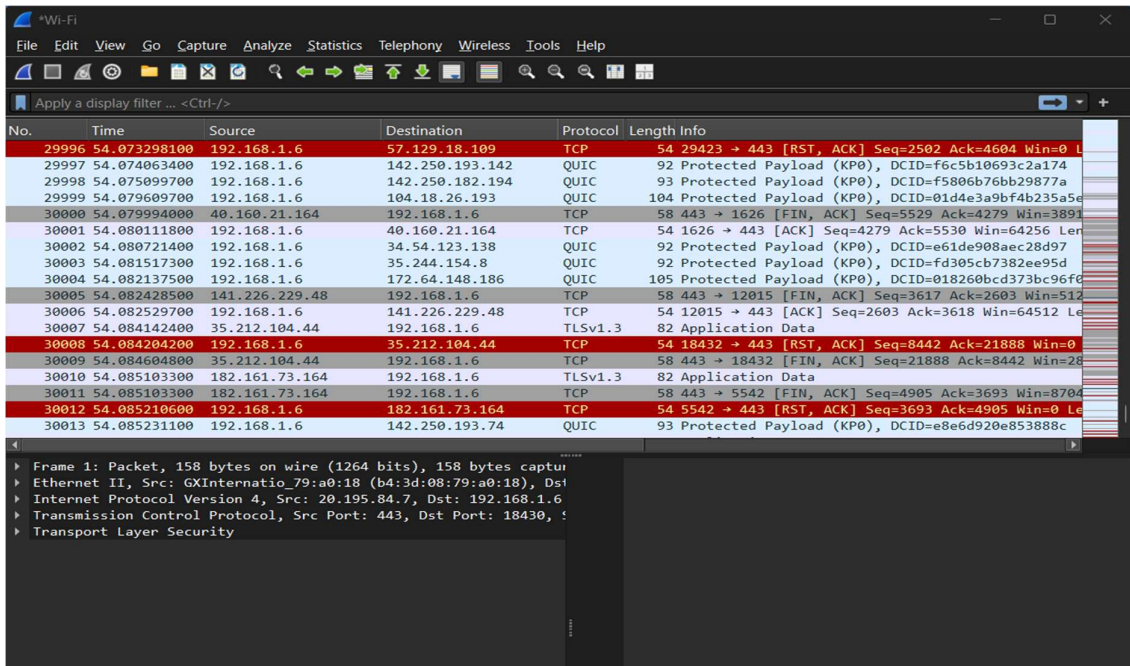
NirSoft Freeware, <https://www.nirsoft.net>

| Properties       |                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL:             | <a href="https://agent.kotak811.com/customer/super_811/details_verification/?token=87734341-35a3-45ef-ac89-637385fe5730">https://agent.kotak811.com/customer/super_811/details_verification/?token=87734341-35a3-45ef-ac89-637385fe5730</a> |
| Title:           | 811Super card details                                                                                                                                                                                                                       |
| Visit Time:      | 09-03-2026 14:17:04                                                                                                                                                                                                                         |
| Visit Count:     | 1                                                                                                                                                                                                                                           |
| Visited From:    |                                                                                                                                                                                                                                             |
| Visit Type:      |                                                                                                                                                                                                                                             |
| Visit Duration:  |                                                                                                                                                                                                                                             |
| Web Browser:     | Edge (Chromium-based)                                                                                                                                                                                                                       |
| User Profile:    | HP                                                                                                                                                                                                                                          |
| Browser Profile: | Default                                                                                                                                                                                                                                     |
| URL Length:      | 110                                                                                                                                                                                                                                         |
| Typed Count:     |                                                                                                                                                                                                                                             |
| History File:    | C:\Users\HP\AppData\Local\Microsoft\Edge\User Data\Default\History                                                                                                                                                                          |
| Record ID:       | 5288                                                                                                                                                                                                                                        |
| OK               |                                                                                                                                                                                                                                             |



## Network Traffic Monitoring and Dashboard Visualization

Wireshark was used to capture live network packets. DNS traffic was filtered to observe domain name resolution queries generated during web browsing. Packet details were analyzed to understand real-time network communication patterns.



\*Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

| No. | Time        | Source      | Destination | Protocol | Length | Info                                              |
|-----|-------------|-------------|-------------|----------|--------|---------------------------------------------------|
| 9   | 4.358092100 | 192.168.1.6 | 192.168.1.1 | DNS      | 89     | Standard query 0x4a93 HTTPS clientservices.google |
| 10  | 4.358421500 | 192.168.1.6 | 192.168.1.1 | DNS      | 89     | Standard query 0x455c A clientservices.googleapi  |
| 11  | 4.368681800 | 192.168.1.1 | 192.168.1.6 | DNS      | 162    | Standard query response 0x4a93 HTTPS clientservi  |
| 12  | 4.369466700 | 192.168.1.1 | 192.168.1.6 | DNS      | 153    | Standard query response 0x455c A clientservices.  |
| 20  | 4.381372300 | 192.168.1.6 | 192.168.1.1 | DNS      | 94     | Standard query 0x1f6c HTTPS oauthaccountmanager.  |
| 21  | 4.381991200 | 192.168.1.6 | 192.168.1.1 | DNS      | 94     | Standard query 0x3e8d A oauthaccountmanager.goog  |
| 24  | 4.382962400 | 192.168.1.6 | 192.168.1.1 | DNS      | 74     | Standard query 0x30af HTTPS www.google.com        |
| 25  | 4.383200600 | 192.168.1.6 | 192.168.1.1 | DNS      | 74     | Standard query 0x2fd2 A www.google.com            |
| 32  | 4.387832800 | 192.168.1.1 | 192.168.1.6 | DNS      | 167    | Standard query response 0x1f6c HTTPS oauthaccoun  |
| 33  | 4.392161500 | 192.168.1.6 | 192.168.1.1 | DNS      | 79     | Standard query 0xeea2 HTTPS accounts.google.com   |
| 34  | 4.392812000 | 192.168.1.6 | 192.168.1.1 | DNS      | 79     | Standard query 0x3773 A accounts.google.com       |
| 35  | 4.393591100 | 192.168.1.1 | 192.168.1.6 | DNS      | 158    | Standard query response 0x3e8d A oauthaccountman  |
| 36  | 4.393591100 | 192.168.1.1 | 192.168.1.6 | DNS      | 147    | Standard query response 0x30af HTTPS www.google.  |
| 38  | 4.394214900 | 192.168.1.1 | 192.168.1.6 | DNS      | 202    | Standard query response 0x2fd2 A www.google.com   |
| 40  | 4.399698500 | 192.168.1.1 | 192.168.1.6 | DNS      | 152    | Standard query response 0xeea2 HTTPS accounts.gc  |
| 41  | 4.399698500 | 192.168.1.1 | 192.168.1.6 | DNS      | 95     | Standard query response 0x3773 A accounts.google  |
| 313 | 4.695958000 | 192.168.1.6 | 192.168.1.1 | DNS      | 88     | Standard query 0x20e1 HTTPS ogads-pa.clients6.g   |
| 314 | 4.696325800 | 192.168.1.6 | 192.168.1.1 | DNS      | 88     | Standard query 0x4768 A ogads-pa.clients6.google  |

Frame 9: Packet, 89 bytes on wire (712 bits), 89 bytes captured  
Internet Protocol Version 4, Src: 192.168.1.6, Dst: 192.168.1.1  
User Datagram Protocol, Src Port: 53166, Dst Port: 53  
Domain Name System (query)

0000 b4 3d 08 79 a0 18 a8 59 5f 2a 28 85 08 00 45 00  
0010 00 4b 2c 75 00 00 80 11 00 00 c0 a8 01 06 c0 a8  
0020 01 01 cf ae 00 35 00 37 83 a0 4a 93 01 00 00 01  
0030 00 00 00 00 00 00 0e 63 6c 69 65 6e 74 73 65 72  
0040 76 69 63 65 73 0a 67 6f 6f 67 6c 65 61 70 69 73  
0050 03 63 6f 6d 00 00 41 00 01

Wireshark - Packet 33 - Wi-Fi

Frame 33: Packet, 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface \Device\NPF\_{D6AED6E2-9EC0-47AB-B6CD-B81493C50C10}, id 0  
Ethernet II, Src: Intel\_2a:28:85 (a8:59:5f:2a:28:85), Dst: GxInt  
Internet Protocol Version 4, Src: 192.168.1.6, Dst: 192.168.1.1  
User Datagram Protocol, Src Port: 56100, Dst Port: 53  
Domain Name System (query)

0000 b4 3d 08 79 a0 18 a8 59 5f 2a 28 85 08 00 45 00  
0010 00 4b 2c 75 00 00 80 11 00 00 c0 a8 01 06 c0 a8  
0020 01 01 db 24 00 35 00 2d 83 96 ea a2 01 00 00 01  
0030 00 00 00 00 00 00 08 61 63 63 6f 75 6e 74 73 06  
0040 67 6f 6f 67 6c 65 03 63 6f 6d 00 00 41 00 01

\*Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

| No.   | Time         | Source         | Destination     | Protocol | Length | Info                                                   |
|-------|--------------|----------------|-----------------|----------|--------|--------------------------------------------------------|
| 29996 | 54.073298100 | 192.168.1.6    | 57.129.18.109   | TCP      | 54     | 29423 → 443 [RST, ACK] Seq=2502 Ack=4604 Win=0 Len=0   |
| 29997 | 54.074063400 | 192.168.1.6    | 142.250.193.142 | QUIC     | 92     | Protected Payload (KP0), DCID=f6c5b10693c2a174         |
| 29998 | 54.075099700 | 192.168.1.6    | 142.250.182.194 | QUIC     | 93     | Protected Payload (KP0), DCID=f5806b76bb29877a         |
| 29999 | 54.079609700 | 192.168.1.6    | 104.18.26.193   | QUIC     | 104    | Protected Payload (KP0), DCID=01d4e3a9bf4b235a5e       |
| 30000 | 54.079994000 | 40.160.21.164  | 192.168.1.6     | TCP      | 58     | 443 → 1626 [FIN, ACK] Seq=5529 Ack=4279 Win=3891 Len=0 |
| 30001 | 54.080111800 | 192.168.1.6    | 40.160.21.164   | TCP      | 54     | 1626 → 443 [ACK] Seq=4279 Ack=5530 Win=64256 Len=0     |
| 30002 | 54.080721400 | 192.168.1.6    | 34.54.123.138   | QUIC     | 92     | Protected Payload (KP0), DCID=e61de908aec28d97         |
| 30003 | 54.081517300 | 192.168.1.6    | 35.244.154.8    | QUIC     | 92     | Protected Payload (KP0), DCID=fd305cb7382ee95d         |
| 30004 | 54.082137500 | 192.168.1.6    | 172.64.148.186  | QUIC     | 105    | Protected Payload (KP0), DCID=018260bcd373bc96f0       |
| 30005 | 54.082428500 | 141.226.229.48 | 192.168.1.6     | TCP      | 58     | 443 → 12015 [FIN, ACK] Seq=3617 Ack=2603 Win=512 Len=0 |
| 30006 | 54.082529700 | 192.168.1.6    | 141.226.229.48  | TCP      | 54     | 12015 → 443 [ACK] Seq=2603 Ack=3618 Win=64512 Len=0    |
| 30007 | 54.084142400 | 35.212.104.44  | 192.168.1.6     | TLSv1.3  | 82     | Application Data                                       |
| 30008 | 54.084204200 | 192.168.1.6    | 35.212.104.44   | TCP      | 54     | 18432 → 443 [RST, ACK] Seq=8442 Ack=21888 Win=0 Len=0  |
| 30009 | 54.084604800 | 35.212.104.44  | 192.168.1.6     | TCP      | 58     | 443 → 18432 [FIN, ACK] Seq=21888 Ack=8442 Win=28 Len=0 |
| 30010 | 54.085103300 | 182.161.73.164 | 192.168.1.6     | TLSv1.3  | 82     | Application Data                                       |
| 30011 | 54.085103300 | 182.161.73.164 | 192.168.1.6     | TCP      | 58     | 443 → 5542 [FIN, ACK] Seq=4905 Ack=3693 Win=8704 Len=0 |
| 30012 | 54.085210600 | 192.168.1.6    | 182.161.73.164  | TCP      | 54     | 5542 → 443 [RST, ACK] Seq=3693 Ack=4905 Win=0 Len=0    |
| 30013 | 54.085231100 | 192.168.1.6    | 142.250.193.74  | QUIC     | 93     | Protected Payload (KP0), DCID=e8e6d920e853888c         |

Frame 1: Packet, 158 bytes on wire (1264 bits), 158 bytes captured on interface Wi-Fi, 158 bytes from 192.168.1.6 to 192.168.1.6

Ethernet II, Src: GXInternatio\_79:a0:18 (b4:3d:08:79:a0:18), Dst: 192.168.1.6

Internet Protocol Version 4, Src: 20.195.84.7, Dst: 192.168.1.6

Transmission Control Protocol, Src Port: 443, Dst Port: 18430, Seq: 2502, Ack: 4604, Win: 0, Len: 0

Transport Layer Security

\*Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

| No. | Time        | Source      | Destination | Protocol | Length | Info                                                                |
|-----|-------------|-------------|-------------|----------|--------|---------------------------------------------------------------------|
| 9   | 4.358092100 | 192.168.1.6 | 192.168.1.1 | DNS      | 89     | Standard query 0x4a93 HTTPS clientservices.google.com               |
| 10  | 4.358421500 | 192.168.1.6 | 192.168.1.1 | DNS      | 89     | Standard query 0x455c A clientservices.google.com                   |
| 11  | 4.368681800 | 192.168.1.1 | 192.168.1.6 | DNS      | 162    | Standard query response 0x4a93 HTTPS clientservices.google.com      |
| 12  | 4.369466700 | 192.168.1.1 | 192.168.1.6 | DNS      | 153    | Standard query response 0x455c A clientservices.google.com          |
| 20  | 4.381372300 | 192.168.1.6 | 192.168.1.1 | DNS      | 94     | Standard query 0x1f6c HTTPS oauthaccountmanager.google.com          |
| 21  | 4.381991200 | 192.168.1.6 | 192.168.1.1 | DNS      | 94     | Standard query 0x3e8d A oauthaccountmanager.google.com              |
| 24  | 4.382962400 | 192.168.1.6 | 192.168.1.1 | DNS      | 74     | Standard query 0x30af HTTPS www.google.com                          |
| 25  | 4.383200600 | 192.168.1.6 | 192.168.1.1 | DNS      | 74     | Standard query 0x2fd2 A www.google.com                              |
| 32  | 4.387832800 | 192.168.1.1 | 192.168.1.6 | DNS      | 167    | Standard query response 0x1f6c HTTPS oauthaccountmanager.google.com |
| 33  | 4.392161500 | 192.168.1.6 | 192.168.1.1 | DNS      | 79     | Standard query 0xea2 HTTPS accounts.google.com                      |
| 34  | 4.392812000 | 192.168.1.6 | 192.168.1.1 | DNS      | 79     | Standard query 0x3773 A accounts.google.com                         |
| 35  | 4.393591100 | 192.168.1.1 | 192.168.1.6 | DNS      | 158    | Standard query response 0x3e8d A oauthaccountmanager.google.com     |
| 36  | 4.393591100 | 192.168.1.1 | 192.168.1.6 | DNS      | 147    | Standard query response 0x30af HTTPS www.google.com                 |
| 38  | 4.394214900 | 192.168.1.1 | 192.168.1.6 | DNS      | 202    | Standard query response 0x2fd2 A www.google.com                     |
| 40  | 4.399698500 | 192.168.1.1 | 192.168.1.6 | DNS      | 152    | Standard query response 0xea2 HTTPS accounts.google.com             |
| 41  | 4.399698500 | 192.168.1.1 | 192.168.1.6 | DNS      | 95     | Standard query response 0x3773 A accounts.google.com                |
| 313 | 4.695958000 | 192.168.1.6 | 192.168.1.1 | DNS      | 88     | Standard query 0x20e1 HTTPS ogads-pa.clients6.google.com            |
| 314 | 4.696325800 | 192.168.1.6 | 192.168.1.1 | DNS      | 88     | Standard query 0x4768 A ogads-pa.clients6.google.com                |

Frame 9: Packet, 89 bytes on wire (712 bits), 89 bytes captured on interface Wi-Fi, 89 bytes from 192.168.1.6 to 192.168.1.1

Ethernet II, Src: Intel\_2a:28:85 (a8:59:5f:2a:28:85), Dst: GXInternatio\_79:a0:18 (b4:3d:08:79:a0:18)

Internet Protocol Version 4, Src: 192.168.1.6, Dst: 192.168.1.1

User Datagram Protocol, Src Port: 53166, Dst Port: 53

Domain Name System (query)

0000 b4 3d 08 79 a0 18 a8 59 5f 2a 28 85 08 00 45 00 ... y...Y

0010 00 4b 2c 75 00 00 80 11 00 00 c0 a8 01 06 c0 a8 ... K,u...

0020 01 01 cf ae 00 35 00 37 83 a0 4a 93 01 00 00 01 ... ..5.7

0030 00 00 00 00 00 0e 63 6c 69 65 6e 74 73 65 72 ... ..c

0040 76 69 63 65 73 0a 67 6f 6f 67 6c 65 61 70 69 73 vices:go

0050 03 63 6f 6d 00 00 41 00 01 ..com..A

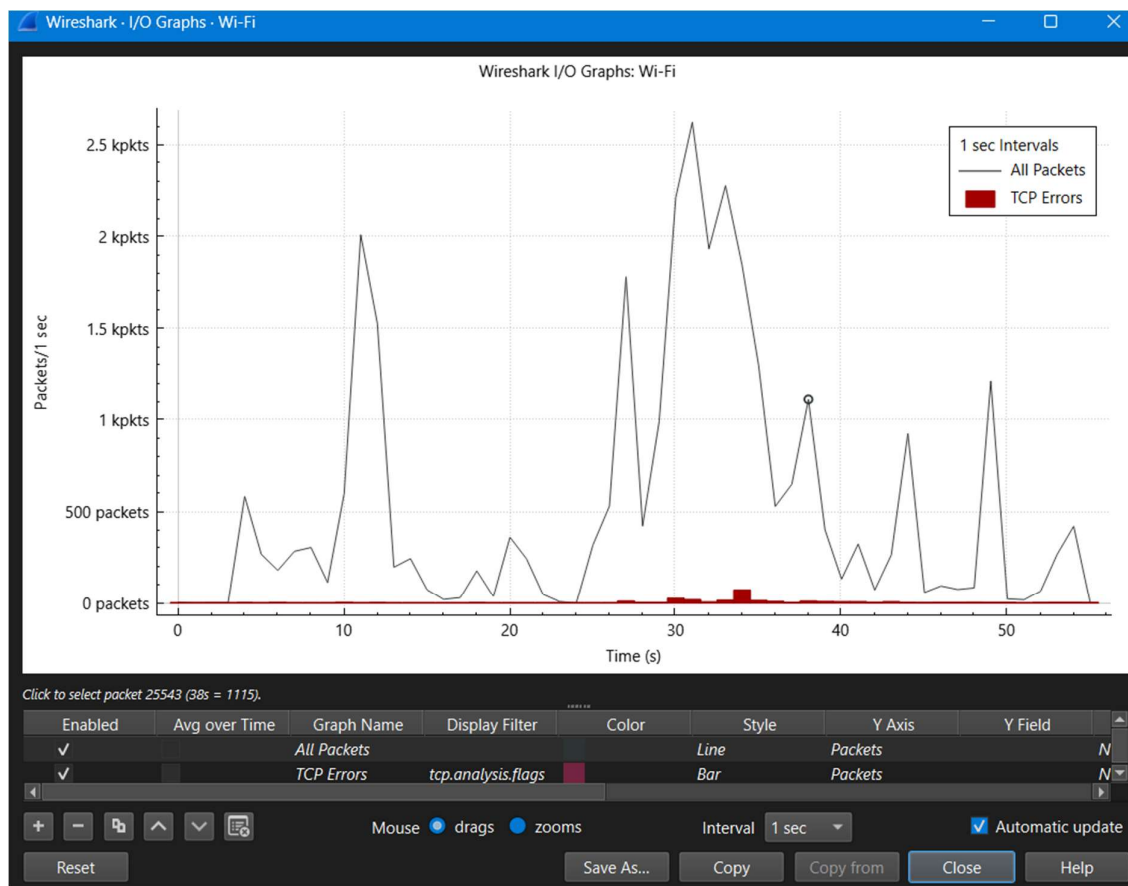
Wireshark · Protocol Hierarchy Statistics · Wi-Fi

| Protocol                          | Percent Packets | Packets | Percent Bytes | Bytes    | Bits/s | End Packets | End Bytes | End Bits/s | PDU's |
|-----------------------------------|-----------------|---------|---------------|----------|--------|-------------|-----------|------------|-------|
| ▼ Frame                           | 100.0           | 30369   | 100.0         | 16101409 | 2310 k | 0           | 0         | 0          | 30369 |
| ▼ Ethernet                        | 100.0           | 30369   | 2.6           | 425250   | 61 k   | 0           | 0         | 0          | 30369 |
| ▼ Internet Protocol Version 6     | 0.0             | 7       | 0.0           | 280      | 40     | 0           | 0         | 0          | 7     |
| ▼ User Datagram Protocol          | 0.0             | 7       | 0.0           | 56       | 8      | 0           | 0         | 0          | 7     |
| ▼ Multicast Domain Name System    | 0.0             | 7       | 0.0           | 343      | 49     | 7           | 343       | 49         | 7     |
| ▼ Internet Protocol Version 4     | 100.0           | 30360   | 3.8           | 607200   | 87 k   | 0           | 0         | 0          | 30360 |
| ▼ User Datagram Protocol          | 46.4            | 14105   | 0.7           | 112840   | 16 k   | 0           | 0         | 0          | 14105 |
| Simple Service Discovery Protocol | 0.0             | 1       | 0.0           | 125      | 17     | 1           | 125       | 17         | 1     |
| QUIC IETF                         | 45.2            | 13735   | 63.8          | 10272145 | 1473 k | 13735       | 10191595  | 1462 k     | 13979 |
| Multicast Domain Name System      | 0.0             | 7       | 0.0           | 343      | 49     | 7           | 343       | 49         | 7     |
| Domain Name System                | 1.2             | 362     | 0.2           | 27150    | 3895   | 362         | 27150     | 3895       | 362   |
| ▼ Transmission Control Protocol   | 53.5            | 16233   | 2.2           | 347300   | 49 k   | 11854       | 259720    | 37 k       | 16233 |
| Transport Layer Security          | 10.8            | 3282    | 27.3          | 4401708  | 631 k  | 3282        | 3756074   | 538 k      | 3438  |
| ▼ Hypertext Transfer Protocol     | 0.0             | 10      | 0.0           | 4037     | 579    | 9           | 2247      | 322        | 10    |
| Media Type                        | 0.0             | 1       | 0.0           | 1236     | 177    | 1           | 1236      | 177        | 1     |
| Domain Name System                | 3.6             | 1086    | 0.5           | 81415    | 11 k   | 1086        | 81415     | 11 k       | 1086  |
| Data                              | 0.0             | 1       | 0.0           | 1        | 0      | 1           | 1         | 0          | 1     |
| Internet Control Message Protocol | 0.1             | 22      | 0.0           | 5008     | 718    | 22          | 5008      | 718        | 22    |
| Address Resolution Protocol       | 0.0             | 2       | 0.0           | 56       | 8      | 2           | 56        | 8          | 2     |
| 802.1Q Virtual LAN                | 11.6            | 3520    | 0.1           | 14104    | 2023   | 0           | 0         | 0          | 3520  |

| Ethernet · 4  |                | IPv4 · 214 |          | IPv6 · 2  |               | TCP · 946   |               | UDP · 295   |  |
|---------------|----------------|------------|----------|-----------|---------------|-------------|---------------|-------------|--|
| Address A     | Address B      | Packets    | Bytes    | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A |  |
| 20.195.84.7   | 192.168.1.6    | 133        | 70 kB    | 0         | 81            | 62 kB       |               |             |  |
| 52.105.243.25 | 192.168.1.6    | 30         | 21 kB    | 39        | 17            | 14 kB       |               |             |  |
| 52.123.128.14 | 192.168.1.6    | 1          | 54 bytes | 186       | 1             | 54 bytes    |               |             |  |
| 192.168.1.6   | 3.33.241.113   | 37         | 17 kB    | 60        | 20            | 13 kB       |               |             |  |
| 192.168.1.6   | 3.112.152.219  | 23         | 7 kB     | 105       | 12            | 3 kB        |               |             |  |
| 192.168.1.6   | 3.121.132.146  | 25         | 9 kB     | 95        | 12            | 3 kB        |               |             |  |
| 192.168.1.6   | 3.150.217.215  | 75         | 29 kB    | 75        | 34            | 8 kB        |               |             |  |
| 192.168.1.6   | 3.209.6.247    | 18         | 9 kB     | 130       | 10            | 3 kB        |               |             |  |
| 192.168.1.6   | 5.181.53.150   | 37         | 16 kB    | 177       | 18            | 9 kB        |               |             |  |
| 192.168.1.6   | 8.2.109.251    | 80         | 31 kB    | 146       | 43            | 14 kB       |               |             |  |
| 192.168.1.6   | 8.2.110.97     | 32         | 13 kB    | 137       | 16            | 6 kB        |               |             |  |
| 192.168.1.6   | 8.214.60.171   | 35         | 11 kB    | 149       | 18            | 5 kB        |               |             |  |
| 192.168.1.6   | 13.114.251.54  | 38         | 13 kB    | 155       | 19            | 5 kB        |               |             |  |
| 192.168.1.6   | 13.159.153.191 | 42         | 13 kB    | 127       | 23            | 5 kB        |               |             |  |
| 192.168.1.6   | 13.213.208.154 | 27         | 9 kB     | 189       | 13            | 3 kB        |               |             |  |
| 192.168.1.6   | 13.214.211.90  | 29         | 10 kB    | 88        | 16            | 4 kB        |               |             |  |
| 192.168.1.6   | 13.225.103.54  | 21         | 13 kB    | 171       | 11            | 5 kB        |               |             |  |
| 192.168.1.6   | 13.225.103.115 | 68         | 47 kB    | 100       | 30            | 8 kB        |               |             |  |
| 192.168.1.6   | 13.228.29.185  | 75         | 22 kB    | 115       | 37            | 10 kB       |               |             |  |
| 192.168.1.6   | 13.250.190.209 | 34         | 10 kB    | 123       | 16            | 4 kB        |               |             |  |
| 192.168.1.6   | 18.66.63.94    | 47         | 24 kB    | 67        | 23            | 8 kB        |               |             |  |
| 192.168.1.6   | 18.66.78.120   | 37         | 14 kB    | 62        | 20            | 4 kB        |               |             |  |
| 192.168.1.6   | 18.139.40.15   | 28         | 11 kB    | 80        | 11            | 3 kB        |               |             |  |
| 192.168.1.6   | 18.153.186.195 | 27         | 9 kB     | 184       | 13            | 3 kB        |               |             |  |
| 192.168.1.6   | 18.164.188.81  | 29         | 11 kB    | 172       | 14            | 3 kB        |               |             |  |
| 192.168.1.6   | 18.164.198.11  | 95         | 39 kB    | 64        | 47            | 21 kB       |               |             |  |
| 192.168.1.6   | 18.164.237.79  | 30         | 11 kB    | 179       | 14            | 3 kB        |               |             |  |
| 192.168.1.6   | 18.215.23.229  | 27         | 9 kB     | 178       | 13            | 3 kB        |               |             |  |
| 192.168.1.6   | 18.245.187.58  | 27         | 15 kB    | 116       | 14            | 6 kB        |               |             |  |

Close

Help



### 3. Security Incident Documentation

| Date       | Source IP | Event ID | Description                                                                       | Action Taken                                                                                |
|------------|-----------|----------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 18-03-2026 | 127.0.0.1 | 4625     | Multiple failed login attempts detected indicating possible brute-force activity. | Logs were monitored and suspicious authentication events were documented for investigation. |

### 4. Alert Rule Design

A detection rule was designed to identify repeated failed login attempts. If five or more failed authentication events (Event ID 4625) occur within five minutes, the SIEM system generates a high-severity alert. This enables SOC analysts to investigate potential unauthorized access attempts and take preventive measures.

Detection Flow:

User Login Attempts → Failed Event Generated → SIEM Counts Events → Condition Met → Alert Triggered → Analyst Investigation

---

## **5. Result**

The practical task provided hands-on experience in monitoring security logs, analyzing browser activity, and capturing network traffic. The activities helped in understanding real-world SOC operations and incident handling procedures.

---

## **6. Conclusion**

This task enhanced practical knowledge of security monitoring techniques and SOC workflows. By performing log analysis, network monitoring, and documentation, essential cybersecurity investigation skills were developed.

---

## **7. Feedback**

The task was highly informative and provided valuable practical exposure to real-time security monitoring tools and incident response concepts.